UC Berkeley

Department of Electrical Engineering and Computer Sciences

EECS 127 - EECS 227AT. Optimization Models in Engineering

Lecture Note 24

Fall 2025

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Definition 1 (Certificate)**.** Consider the following primal optimization problem:

$$\begin{aligned} \min \quad & 0, \\ \text{s.t.} \quad & f_i(\boldsymbol{x}) \le 0,\ i = 1, \dots, k, \\ & h_j(\boldsymbol{x}) = 0,\ j = 1, \dots, m, \end{aligned}$$

and its dual problem:

$$\begin{aligned} \max \quad & d(\boldsymbol{\lambda}, \boldsymbol{\mu}), \\ \text{s.t.} \quad & \boldsymbol{\lambda} \ge \mathbf{0}. \end{aligned}$$

If we can find a pair $(\hat{\boldsymbol{\lambda}}, \hat{\boldsymbol{\mu}})$ with $\hat{\boldsymbol{\lambda}} \ge \mathbf{0}$ such that $d(\hat{\boldsymbol{\lambda}}, \hat{\boldsymbol{\mu}}) > 0$, then

$$p^* \ge d^* \ge d(\hat{\boldsymbol{\lambda}}, \hat{\boldsymbol{\mu}}) > 0 \ \Rightarrow\ p^* \ne 0 \ \Rightarrow\ p^* = +\infty.$$

Therefore, finding this single point certifies that

$$\begin{cases} f_i(\boldsymbol{x}) \le 0,\ i = 1, \dots, k, \\ h_j(\boldsymbol{x}) = 0,\ j = 1, \dots, m \end{cases}$$

has no solution. Such point $(\hat{\boldsymbol{\lambda}}, \hat{\boldsymbol{\mu}})$ is called a *certificate*.

Example 1 (Certificate)**.** We prove that there is no $\boldsymbol{x} \in \mathbb{R}^3$ such that

$$\begin{cases} x_1^2 + x_2^2 + x_3^3 = 1, \\ x_1 + 2x_2 + 3x_3 \ge 5. \end{cases} \tag{1}$$

Define the primal problem:

$$\begin{aligned} \min \quad & 0, \\ \text{s.t.} \quad & x_1^2 + x_2^2 + x_3^2 = 1, \\ & x_1 + 2x_2 + 3x_3 \ge 5. \end{aligned}$$

Let λ, μ be the Lagrange multipliers. The Lagrangian is

$$\begin{aligned} L(\boldsymbol{x}, \lambda, \mu) &= 0 + \mu(x_1^2 + x_2^2 + x_3^2 - 1) + \lambda(5 - x_1 - 2x_2 - 3x_3) \\ &= (\mu x_1^2 - \lambda x_1) + (\mu x_2^2 - 2\lambda x_2) + (\mu x_3^2 - 3\lambda x_3) - \mu + 5\lambda. \end{aligned}$$

The objective is separable in x_1, x_2, x_3:

$$\min_{\boldsymbol{x}} L(\boldsymbol{x}, \lambda, \mu) \quad \Longleftrightarrow \quad \min_{x_i} L(\boldsymbol{x}, \lambda, \mu),\ i = 1, 2, 3.$$

For x_1, we have

$$\min_{x_1} \mu x_1^2 - \lambda x_1 = \begin{cases} -\infty, & \text{if } \mu < 0, \\ -\infty, & \text{if } \mu = 0, \lambda \ne 0, \\ 0, & \text{if } \mu = 0, \lambda = 0, \\ -\lambda^2/4\mu, & \text{if } \mu > 0, \end{cases}$$

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where the last case is because $\mu x_1^2 - \lambda x_1$ is convex when $\mu > 0$, and

$$\frac{\partial(\mu x_1^2 - \lambda x_1)}{\partial x_1} = 0 \quad \Rightarrow \quad x_1^*(\lambda, \mu) = \frac{\lambda}{2\mu}.$$

Similarly, when $\mu > 0$, we have

$$x_2^*(\lambda, \mu) = \frac{2\lambda}{2\mu}, \quad x_3^*(\lambda, \mu) = \frac{3\lambda}{2\mu}.$$

The dual function is

$$\begin{aligned} d(\lambda, \mu) &= \mu \left(\left(\frac{\lambda}{2\mu}\right)^2 + \left(\frac{2\lambda}{2\mu}\right)^2 + \left(\frac{3\lambda}{2\mu}\right)^2 - 1 \right) + \lambda \left(5 - \left(\frac{\lambda}{2\mu}\right) - 2\left(\frac{2\lambda}{2\mu}\right) - 3\left(\frac{3\lambda}{2\mu}\right) \right) \\ &= \frac{\lambda^2}{\mu}\left(\frac{1}{4} + 1 + \frac{9}{4}\right) - \mu + \frac{\lambda^2}{\mu}\left(-\frac{1}{2} - 2 - \frac{9}{2}\right) + 5\lambda \\ &= -3.5\frac{\lambda^2}{\mu} - \mu + 5\lambda. \end{aligned}$$

Select $\hat{\lambda} = \hat{\mu} = 1$, then $d(1, 1) = -3.5 - 1 + 5 > 0$. Therefore, $(1, 1)$ certifies that the equations (1) has no solution. ■

- In the non-convex case, we have $p^* \geq d^*$, and when $p^* = +\infty$ (infeasible) but $d^* < 0$, our certification method does not work (as the desired $(\hat{\boldsymbol{\lambda}}, \hat{\boldsymbol{\mu}})$ doesn't exist). So, it is useful only if $d^* > 0$.
- Certifying that a set of non-convex equations has no solution is NP-hard in general. So, we can't solve it efficiently all the time.
- However, in the convex case, we have $p^* = d^*$ if the dual problem satisfies the Slater's condition.[1] (Note: the primal problem is not supposed to satisfy the Slater's condition since we aim to prove that it is infeasible.)
- In the linear case, the *Farkas' lemma* is used to prove that a set of linear equalities and inequalities has no solution.
- Rather than taking the dual directly, let's first introduce slack variables and write the linear constraints in standard form:

  $$\mathbf{A}\boldsymbol{x} = \boldsymbol{b}, \quad \boldsymbol{x} \geq \mathbf{0}.$$

  Consider the primal problem:

  $$\begin{aligned} \min_{\boldsymbol{x} \in \mathbb{R}^n} \quad & 0, \\ \text{s.t.} \quad & \mathbf{A}\boldsymbol{x} = \boldsymbol{b}, \\ & \boldsymbol{x} \geq \mathbf{0}. \end{aligned}$$

[1] You can prove this by Applying Theorem 2 in Lecture Note 23 to the dual problem.

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Let the rows of $\mathbf{A}$ be $\boldsymbol{a}_1^{\mathrm{T}}, \ldots, \boldsymbol{a}_m^{\mathrm{T}}$. The Lagrangian is

$$\begin{aligned} L(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{\mu}) &= 0 + \sum_{j=1}^{m} \mu_j \left(\boldsymbol{a}_j^{\mathrm{T}} \boldsymbol{x} - b_j\right) - \sum_{i=1}^{n} \lambda_i x_i \\ &= \left(\sum_{j=1}^{m} \mu_j \boldsymbol{a}_j - \boldsymbol{\lambda}\right)^{\mathrm{T}} \boldsymbol{x} - \sum_{j=1}^{m} \mu_j b_j. \end{aligned}$$

Since this is an affine function of $\boldsymbol{x}$, we have

$$d(\boldsymbol{\lambda}, \boldsymbol{\mu}) = \begin{cases} -\sum_{j=1}^{m} \mu_j b_j, & \text{if } \sum_{j=1}^{m} \mu_j \boldsymbol{a}_j - \boldsymbol{\lambda} = \mathbf{0}, \\ -\infty, & \text{otherwise.} \end{cases}$$

The dual problem can be formulated as

$$\begin{aligned} \max_{\boldsymbol{\lambda}, \boldsymbol{\mu}} \quad & -\underbrace{\boxed{\sum_{j=1}^{m} \mu_j b_j}}_{\boldsymbol{b}^{\mathrm{T}}\boldsymbol{\mu}}, \\ \text{s.t.} \quad & \underbrace{\boxed{\sum_{j=1}^{m} \mu_j \boldsymbol{a}_j}}_{\mathbf{A}^{\mathrm{T}}\boldsymbol{\mu}} - \boldsymbol{\lambda} = \mathbf{0}, \\ & \boldsymbol{\lambda} \geq \mathbf{0}. \end{aligned} \tag{2}$$

- Notice that $\boldsymbol{\mu} = \mathbf{0}, \boldsymbol{\lambda} = \mathbf{0}$ are feasible for (2), thus the Slater's condition holds for (2), and we have $p^* = d^*$.
- So, $p^* \neq 0$ iff $p^* = +\infty$ or $d^* = +\infty$.
- To have $p^* = +\infty$, it is necessary and sufficient to have a pair $(\hat{\boldsymbol{\lambda}}, \hat{\boldsymbol{\mu}})$ such that

$$\mathbf{A}^{\mathrm{T}} \hat{\boldsymbol{\mu}} - \hat{\boldsymbol{\lambda}} = \mathbf{0}, \quad \hat{\boldsymbol{\lambda}} \geq \mathbf{0}, \quad -\boldsymbol{b}^{\mathrm{T}} \hat{\boldsymbol{\mu}} > 0,$$

or equivalently,

$$\mathbf{A}^{\mathrm{T}} \hat{\boldsymbol{\mu}} \geq \mathbf{0}, \quad -\boldsymbol{b}^{\mathrm{T}} \hat{\boldsymbol{\mu}} > \mathbf{0}.$$

Theorem 1 (Farkas' lemma)**.** *The equations $\mathbf{A}\boldsymbol{x} = \boldsymbol{b}$, $\boldsymbol{x} \geq \mathbf{0}$ have no solution iff there exists $\hat{\boldsymbol{\mu}}$ such that $\mathbf{A}^{\mathrm{T}} \hat{\boldsymbol{\mu}} \geq \mathbf{0}$, $\boldsymbol{b}^{\mathrm{T}} \hat{\boldsymbol{\mu}} < \mathbf{0}$.*

Constraint elimination. Consider convex optimization:

$$\begin{aligned} \min \quad & f_0(\boldsymbol{x}), \\ \text{s.t.} \quad & f_i(\boldsymbol{x}) \leq 0, \ i = 1, \ldots, k, \\ & h_j(\boldsymbol{x}) = 0, \ j = 1, \ldots, m. \end{aligned} \tag{3}$$

Let $\boldsymbol{x}^*$ be an arbitrary solution. The constraint $f_i(\boldsymbol{x}) \leq 0$ is called *active* or *binding* at $\boldsymbol{x}^*$ if $f_i(\boldsymbol{x}^*) = 0$. It is called *inactive* or *non-binding* if $f_i(\boldsymbol{x}^*) < 0$.

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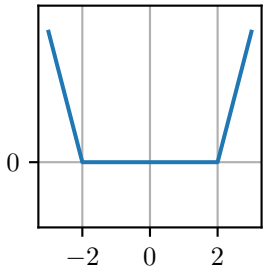
Theorem 2. *Let $\boldsymbol{x}^*$ be a solution to (3). Consider the optimization (4) below where inactive constraints are removed:*

$$\begin{aligned} \min \quad & f_0(\boldsymbol{x}), \\ s.t. \quad & f_i(\boldsymbol{x}) \leq 0, \ i \in \mathcal{A}(\boldsymbol{x}^*), \\ & h_j(\boldsymbol{x}) = 0, \ j = 1, \ldots, m, \end{aligned} \tag{4}$$

where $\mathcal{A}(\boldsymbol{x}^*)$ is the set of indices of active constraints at $\boldsymbol{x}^*$. Then $\boldsymbol{x}^*$ is also a solution to (4), so (4) and (3) have the same optimal objective values.

Note: when we eliminate constraints to obtain (4), new optimal solutions could be created for (4) that are not feasible for (3), but $\boldsymbol{x}^*$ remains an optimal solution.

Example 2 (Constraint elimination)**.** Let $f(x)$ be as follows.



Consider the following optimization:

$$\begin{aligned} \min \quad & f(x), \\ \text{s.t.} \quad & -1 \leq x \leq 1. \end{aligned}$$

Pick $x^* = 0$, then both constraints $-1 \leq x$ and $x \leq 1$ are inactive.
The optimization after constraint elimination is simply $\min f(x)$. Here $x^* = 0$ is still a solution, but $x = 2$ is also a solution that is not feasible to the original problem. ■

- Since $\boldsymbol{x}^*$ is unknown, we don't know which constraints would be active, so we can't reduce (3) to (4) before solving (3). However, (4) implies that inactive constraints are not important.
- Now, assume that the Slater's condition holds so we can use KKT.
- By complementary slackness, $\lambda_i f_i(\boldsymbol{x}^*) = 0$.
- If $f_i(\boldsymbol{x}^*) \leq 0$ is inactive, then $f_i(\boldsymbol{x}^*) < 0 \Rightarrow \lambda_i^* = 0$.
- Thus, $\lambda_i^* = 0$ when the constraint is not important.
- Degenerate case: $\lambda_i^* = f_i(\boldsymbol{x}^*) = 0$. Again, it can be shown that since $\lambda_i^* = 0$, the constraint is not important.
- This will be formalized through sensitivity analysis.

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Sensitivity analysis. Consider the original convex optimization problem:

$$\begin{aligned} \min \quad & f_0(\boldsymbol{x}), \\ \text{s.t.} \quad & f_i(\boldsymbol{x}) \le 0,\ i = 1, \dots, k, \\ & h_j(\boldsymbol{x}) = 0,\ j = 1, \dots, m, \end{aligned}$$

and the perturbed convex optimization problem:

$$\begin{aligned} \min \quad & f_0(\boldsymbol{x}), \\ \text{s.t.} \quad & f_i(\boldsymbol{x}) \le v_i,\ i = 1, \dots, k, \\ & h_j(\boldsymbol{x}) = w_j,\ j = 1, \dots, m, \end{aligned}$$

where v_i's and w_j's are arbitrary scalars. So, the zeros on the right-hand side are perturbed to v_i's and w_j's.

- Let p^* denote the optimal objective value of the original problem.
- Let $p^*(\boldsymbol{v}, \boldsymbol{w})$ denote the optimal objective value of the perturbed problem.
- We have $p^*(\mathbf{0}_k, \mathbf{0}_m) = p^*$, and if for some $\boldsymbol{v}$ and $\boldsymbol{w}$ the perturbed problem is infeasible, then $p^*(\boldsymbol{v}, \boldsymbol{w}) = +\infty$.

Theorem 3. *$p^*(\boldsymbol{v}, \boldsymbol{w})$ is a convex function of $(\boldsymbol{v}, \boldsymbol{w})$.*

Theorem 4. *Assume the Slater's condition holds for the original problem. If $p^*(\boldsymbol{v}, \boldsymbol{w})$ is differentiable at $(\mathbf{0}, \mathbf{0})$, then:*

$$\begin{aligned} \lambda_i^* &= -\frac{\partial p^*}{\partial v_i}(\mathbf{0}, \mathbf{0}), \quad i = 1, \dots, k, \\ \mu_j^* &= -\frac{\partial p^*}{\partial w_j}(\mathbf{0}, \mathbf{0}), \quad j = 1, \dots, m, \end{aligned} \tag{5}$$

where $(\boldsymbol{\lambda}^*, \boldsymbol{\mu}^*)$ is the unique solution of KKT for some $\boldsymbol{x}^*$ (or unique solution of the dual problem).

Note: Differentiability guarantees uniqueness of $(\boldsymbol{\lambda}^*, \boldsymbol{\mu}^*)$. One sufficient condition to have differentiability is that $f_0(\boldsymbol{x})$ being strictly convex and $f_0(\boldsymbol{x}), \dots, f_k(\boldsymbol{x})$ being twice continuously differentiable (so we can calculate their Hessians).
(5) means:

$$p^*(\boldsymbol{v}, \boldsymbol{w}) \simeq \underbrace{p^* - \sum_{i=1}^{k} \lambda_i^* v_i - \sum_{j=1}^{m} \mu_j^* w_j}_{\text{first-order taylor approximation}}.$$

(1) If λ_i^* or μ_j^* equals zero, then changing the constraint a bit won't affect the optimal objective. Thus, these constraints can be eliminated.

(2) If λ_i^* or μ_j^* is small, then the optimization is not sensitive to those constraints.

(3) If λ_i^* or μ_j^* is large, then the optimization is very sensitive to those constraints.

Therefore, the Lagrange multipliers act as sensitivity parameters.

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Matrix optimization. So far, the variable of optimization has been a vector in $\mathbb{R}^n$. What if it is a matrix $\mathbf{X} \in \mathbb{R}^{m\times n}$? Consider

$$\begin{aligned} \min_{\mathbf{X}\in\mathbb{R}^{m\times n}} \quad & f_0(\mathbf{X}), \\ \text{s.t.} \quad & f_i(\mathbf{X}) \le 0,\ i = 1,\dots,k, \\ & h_j(\mathbf{X}) = 0,\ j = 1,\dots,m, \end{aligned}$$

where $f_0,\dots,f_k,h_1,\dots,h_m\colon \mathbb{R}^{m\times n} \to \mathbb{R}$.

- Define $\boldsymbol{x} \in \mathbb{R}^{mn}$ as a vector that includes all entries of $\mathbf{X}$. Then, there exist functions $\tilde{f}_i$ $(i = 1,\dots,k)$ and $\tilde{h}_j$ $(j = 1,\dots,m)$ such that $f_i(\mathbf{X}) = \tilde{f}_i(\boldsymbol{x})$ and $h_j(\mathbf{X}) = \tilde{h}_j(\boldsymbol{x})$.
- So, the optimization can be rewritten into standard form:

$$\begin{aligned} \min_{\boldsymbol{x}\in\mathbb{R}^{mn}} \quad & \tilde{f}_0(\boldsymbol{x}), \\ \text{s.t.} \quad & \tilde{f}_i(\boldsymbol{x}) \le 0,\ i = 1,\dots,k, \\ & \tilde{h}_j(\boldsymbol{x}) = 0,\ j = 1,\dots,m. \end{aligned}$$

- Now, all of our previous theorems about convexity and optimality can be applied to the reformulated problem.

Example 3. Consider the following matrix optimization problem:

$$\begin{aligned} \min_{\mathbf{X}\in\mathbb{R}^{2\times 2}} \quad & \left\| \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \mathbf{X} - \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix} \right\|_2, \\ \text{s.t.} \quad & X_{11} + X_{12} \le 5, \\ & X_{21} + 5X_{22}^2 = 4. \end{aligned}$$

We can write it as an equivalent vector optimization problem:

$$\begin{aligned} \min_{\boldsymbol{x}\in\mathbb{R}^4} \quad & \underbrace{\left\| \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} x_1 & x_2 \\ x_3 & x_4 \end{bmatrix} - \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix} \right\|_2}_{\tilde{f}_0(\boldsymbol{x})}, \\ \text{s.t.} \quad & \underbrace{x_1 + x_2 - 5}_{\tilde{f}_1(\boldsymbol{x})} \le 0, \\ & \underbrace{x_3 + 5x_4^2 - 4}_{\tilde{h}_1(\boldsymbol{x})} = 0. \end{aligned}$$

■

PCA is an example of matrix optimization.

- Given $\mathbf{Y} \in \mathbb{R}^{m\times n}$ and $r \le \min(m,n)$, solve:

$$\begin{aligned} \min_{\mathbf{X}\in\mathbb{R}^{m\times n}} \quad & \|\mathbf{X} - \mathbf{Y}\|_{\mathrm{F}}, \\ \text{s.t.} \quad & \operatorname{rank}(\mathbf{X}) \le r. \end{aligned}$$

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- Since $\text{rank}(\mathbf{X}) \leq r$ is non-standard, we previously wrote $\mathbf{X}$ as $\mathbf{C}\mathbf{D}^{\mathrm{T}}$ where $\mathbf{C} \in \mathbb{R}^{m\times r}$ and $\mathbf{D} \in \mathbb{R}^{n\times r}$, and then the optimization in $\mathbf{X}$ can be rewritten as

$$\min_{\mathbf{C},\mathbf{D}} \quad \|\mathbf{C}\mathbf{D}^{\mathrm{T}} - \mathbf{Y}\|_{\mathrm{F}}.$$

  In this case, we can put it in the standard form

$$\min_{\boldsymbol{x}} \quad f_0(\boldsymbol{x}),$$

  where $\boldsymbol{x}$ includes all entries of $\mathbf{C}$ and $\mathbf{D}$.

- This is unnecessary since we showed that the optimal $\mathbf{C}^*$ and $\mathbf{D}^*$ can be obtained from the SVD of $\mathbf{Y}$.

- However, if there are additional constraints like

$$\begin{aligned} \min_{\mathbf{X}} \quad & \|\mathbf{X} - \mathbf{Y}\|_{\mathrm{F}}, \\ \text{s.t.} \quad & \text{rank}(\mathbf{X}) \leq r, \\ & |X_{ij}| \leq 1, \ \forall i, j, \end{aligned}$$

  then this is a constrained low-rank decomposition, and there is no formula for its solution. We should rewrite it in terms of a vector $\boldsymbol{x}$ and then use optimality conditions (such as KKT) to solve it.

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